DEEPAK KATUKURI

DATA ANALYSIS · ARTIFICIAL INTELLIGENCE · PYTHON AUTOMATION



CONTACT

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TECHNICAL SKILLS

Data Analysis:
Python, Pandas, NumPy, SQL, PL/SQL, Scikit-Learn, data querying & validation

Dashboards & Visualization:
Recharts, Matplotlib, Seaborn, KPI dashboards, ad-hoc reporting

Python Microservices & APIs:
FastAPI, REST API design, Docker, CI/CD, Git/GitHub

AI / LLM Systems:
Retrieval-Augmented Generation, embeddings, vector search, prompt engineering, LLM orchestration (Ollama)

Deep Learning:
PyTorch, TensorFlow, CLIP, DistilBERT, fine-tuning, Hugging Face

Tools:
Git/GitHub, pytest, Jupyter, IBM Watson

EDUCATION

MSc Data Science
University of Europe, Germany
Sep 2025 – Present

BTech – CS & Engineering
MLRIT, India
Jun 2018 – Jul 2022

CERTIFICATIONS

PL/SQL – Database Management

Core Java – Software Development

OneShield – P&C Insurance

LANGUAGES

English – Fluent
German – A1/A2 (learning)

PROFILE

Data Science MSc candidate and former production Software Engineer who turns complex data into decision-ready insight — from Python-based automation pipelines to stakeholder-facing dashboards. Built a framework-free Retrieval-Augmented Generation pipeline and a full-stack analytics dashboard (FastAPI + PostgreSQL + Recharts) from scratch, backed by 2.5 years translating underwriting and claims requirements into validated backend logic at a P&C insurance software vendor. Seeking a Werkstudent role in Datenanalyse & Künstliche Intelligenz at S-Kreditpartner to help turn complex bank data into steering impulses for Risikomanagement and Banksteuerung.

EXPERIENCE

Software Engineer
Mar 2022 – Sep 2024

ValueMomentum

- Queried, maintained, and validated policy-lifecycle data across 2 enterprise P&C insurance and surety-bond applications on OneShield PAS, resolving production defects flagged during data-quality checks.
- Built and optimized PL/SQL stored procedures and functions powering backend insurance data operations and endorsement modules, enforcing business-rule validation logic across policy records.
- Translated business requirements from underwriting and claims stakeholders into technical specs during sprint planning, then validated decision logic through QA/UAT test cases before deployment across the full policy lifecycle.

ML Intern
Jun – Jul 2019

SmartBridge × IBM

- Built a lung-cancer risk-prediction ensemble (Random Forest, KNN, Gradient Boosting, Neural Network) and deployed it via IBM Watson ML, communicating results to non-technical reviewers with Matplotlib/Seaborn visualizations.

PROJECTS

RA Nutrition & Lifestyle App – Full-Stack Analytics Dashboard

React · FastAPI · PostgreSQL · Recharts · personal product build

- Built a Python-based FastAPI + PostgreSQL microservice backend (4 logical schemas) with JWT auth and token refresh, powering a React/Vite/Tailwind frontend end-to-end.
- Shipped a rule-based recommendation engine and a weekly analytics dashboard (Recharts) that closes the loop from raw meal/symptom logs to actionable, decision-ready insight.
- Architected (not yet shipped) an ML forecasting tier — Scikit-Learn/XGBoost and PyTorch LSTM/GRU/TCN — for flare-risk prediction, scoping model validation before building it.

RAG Pipeline – Retrieval-Augmented Generation from Scratch

Python · sentence-transformers · NumPy · Ollama · github.com/Deepak589/RAG_pipeline

- Built a framework-free RAG pipeline in ~270 lines of Python (no LangChain, LlamaIndex, or vector DB), implementing chunking, embedding, similarity search, and prompt assembly to keep every retrieval mechanic explicit and auditable.
- Improved retrieval relevance by pairing a dense bi-encoder retriever (all-MiniLM-L6-v2, cosine top-3) with a hand-rolled TF-IDF baseline and heading-prefix chunking to cut off-topic matches.
- Kept generation crash-free with a built-in quality check: degrading to retrieved-context output whenever the local LLM was offline, instead of failing silently.

Food Image-to-Recipe Retrieval – Multimodal Deep Learning

PyTorch · CLIP ViT-B/32 · DistilBERT · InfoNCE · MSc final project

- Built a dual-encoder retrieval model mapping food images (CLIP ViT-B/32) and recipe text (DistilBERT) into a shared embedding space, reaching a validation median rank of 10/200 on 13.5K images.
- Validated the pipeline with 58 unit tests and recall@k / median-rank evaluation in both retrieval directions before calling it done.